A. Further Details on the Experimental Setup

A.1. Model Specification

The following table lists the models and their corresponding links.

Models	Links
`Llama-2-7B` (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-7b-hf
`Llama-2-7B-Chat` (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-7b-chat-hf
`Llama-2-13B-Chat` (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-13b-chat-hf
`Llama-2-70B-Chat` (Touvron et al., 2023b)	https://huggingface.co/meta-llama/Llama-2-70b-chat-hf
`Llama-3-8B` (AI@Meta, 2024)	https://huggingface.co/meta-llama/Meta-Llama-3-8B
`Llama-3-8B-Instruct` (AI@Meta, 2024)	https://huggingface.co/meta-llama/Llama-2-7b-hf
`Llama-3-70B-Instruct` (AI@Meta, 2024)	https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct
`Mistral-7B` (Jiang et al., 2023)	https://huggingface.co/mistralai/Mistral-7B-v0.1
`Mistral-7B-Instruct` (Jiang et al., 2023)	https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2
`Qwen2-7B` (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-7B
`Qwen2-7B-Instruct` (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-7B-Instruct
`Qwen2-72B-Instruct` (Team, 2024a)	https://huggingface.co/Qwen/Qwen2-72B-Instruct
`Qwen2.5-1.5B-Instruct` (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct
`Qwen2.5-32B-Instruct` (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-32B-Instruct
`Qwen2.5-72B-Instruct` (Team, 2024b)	https://huggingface.co/Qwen/Qwen2.5-72B-Instruct

A.2. Dataset Specification

The following table lists the datasets and their corresponding links.

Datasets	Links
`Ultrachat` (Ding et al., 2023)	https://huggingface.co/datasets/stingning/ultrachat
`AlpacaEval` (Taori et al., 2023)	https://huggingface.co/datasets/tatsu-lab/alpaca
`GSM8K` (Cobbe et al., 2021)	https://huggingface.co/datasets/openai/gsm8k
`MATH` (Saxton et al., 2019)	https://huggingface.co/datasets/deepmind/math_dataset
`TinyStories` (Eldan & Li, 2023)	https://huggingface.co/datasets/roneneldan/TinyStories
`ROCStories` (Mostafazadeh et al., 2016)	https://huggingface.co/datasets/Ximing/ROCStories

CommonsenseQA (Talmor et al., 2019)	https://huggingface.co/datasets/tau/commonsense_qa
SocialIQA (Sap et al., 2019)	https://huggingface.co/datasets/allenai/social_i_qa
MedMCQA (Pal et al., 2022)	https://huggingface.co/datasets/openlifescienceai/medmcqa
ARC-Challenge (Clark et al., 2018)	https://huggingface.co/datasets/allenai/ai2_arc
CREAK (Onoe et al., 2021)	https://huggingface.co/datasets/amydeng2000/CREAK
FEVER (Thorne et al., 2018)	https://huggingface.co/datasets/fever/fever

A.3. Detailed Process of Response Collection and Labeling

In this section, we detail the process of collecting a dataset $\mathcal{D} = \{\mathcal{H}_i, \hat{g}_i\}_{i=1}^N$ for each task $T = (p(\mathbf{x}), g(\mathbf{y}))$, pairing prompt representations with their corresponding attribute labels. First, we construct the prompt distribution $p(\mathbf{x})$ to elicit responses with target attributes from the models (Sec.A.3.1). Second, we label these responses according to specific criteria $\hat{g}_i = g(\mathbf{y}_i)$ to capture their key attributes (Sec.A.3.2). Finally, we collect representations $\mathcal{H}_i = \{\mathbf{H}_{\mathbf{x}_i}^l\}_{l=1}^L$ for each prompt (Sec. A.3.3).

A.3.1. Prompt templates

To elicit responses with target attributes, we construct prompt distributions using carefully designed templates paired with datasets. We present the prompt templates for both chat and base models across all tasks, along with representative input-output examples.

Task 1: Response Length

Prompt for fine-tuned models

```
''
{data}
''
```

($\rightarrow$ Gets formatted according to model's template)

Example Response

Data: Why are oceans important to the global ecosystem?
Output: The oceans play a crucial role [...]

Prompt for base models

```
''
Q: How can cross training benefit athletes?
A: Cross training offers various benefits [...]  [END OF RESPONSE]
Q: What role does collaboration play in creativity?
A: Collaboration and originality complement each other [...]  [END
OF RESPONSE]
Q: {data}
A:
''
```

Example Response

Data: What are positive impacts of Reality TV?
Output: Reality TV provides entertainment and [...] [END OF RESPONSE]

Task 2: Reasoning Steps

Prompt for fine-tuned models

```
''
Provide step-by-step solution, starting with 'Step 1:'.
Problem:
{data}
```

''

(→ Gets formatted according to model's template)

Example Response

Data: Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all?
Output: Step 1: Write down the information [...]

Prompt for base models

''

Solve this problem step-by-step, starting with 'Step 1:'.
Few-shot examples:
Problem: Let f(x)={ax+3 if x>2; x-5 if -2≤x≤2; 2x-b if x<-2}.
Find a+b if f is continuous.
Step 1: At x=2: a(2)+3=2-5 [...] [END OF RESPONSE]
Problem: If x=2 and y=5, find (x^4+2y^2)/6.
Step 1: Substitute: (2^4+2(5^2))/6 [...] [END OF RESPONSE]
Problem: {data}

''

Example Response

Data: Weng earns $12 an hour for babysitting. Yesterday, she just did 50 minutes of babysitting. How much did she earn?
Output: Step 1: Substitute: 12(50/60) [...] [END OF RESPONSE]

Task 3: Character Choices

Prompt for fine-tuned models

''

Here's the first sentence of a story: {data}
Continue this story with one sentence that introduces a new animal character.

''

(→ Gets formatted according to model's template)

Example Response

Data: Once upon a time, there was a big car named Dependable.
Output: As Dependable was cruising down the highway, a chatty parrot [...]

Prompt for base models

''

First sentence: Lily was a little mouse who liked to follow her big brother Leo.
Continuation: The garden was peaceful that morning until [...] [Animal: owl] [END OF RESPONSE]
First sentence: Lila and Ben were playing in the park with their toys.
Continuation: While building their epic sandcastle [...] [Animal: rabbit] [END OF RESPONSE]
First sentence: Sara was lonely.
Continuation: As she sat on the front steps drawing patterns [...] [Animal: puppy] [END OF RESPONSE]
First sentence: Lily and Ben were twins who liked to go on walks with their mom and dad.
Continuation: Their morning hike through the woods [...] [Animal: squirrel] [END OF RESPONSE]
First sentence: {data}
Continuation:

''

Example Response

Data: One day, a girl named Mia went for a walk.
Output: As she strolled through the park, she noticed a group of birds [...] [END OF RESPONSE]

Task 4: Multiple-Choice Answers

Prompt for fine-tuned models

''
Before choosing your answer, *briefly explain why in one short sentence*. Then select from the options:
{data}
''

(→ Gets formatted according to model's template)

Example Response

Data: Sammy wanted to go to where the people were. Where might he go? A) race track B) populated areas C) the desert D) apartment E) roadblock
Output: **He wants to be around people, so he would go to a populated area.**B) populated areas

Prompt for base models

''
Select the correct answer. Choose the single best answer.
Q: The sanctions against the school were a punishing blow, and they seemed to what the efforts the school had made to change?
A)ignore B)enforce C)authoritarian D)yell at E)avoid
A: The sanctions ignore the school's reform efforts, contradicting their purpose. Therefore A.[END OF RESPONSE]
Q: Sammy wanted to go to where the people were. Where might he go?
A)race track B)populated areas C)the desert D)apartment E)roadblock
A: If Sammy wants to find people, he would logically go to populated areas where many people gather. Therefore B.[END OF RESPONSE]
Q: {data}
A:
''

Example Response

Data: Where do you put your grapes just before checking out? A) mouth B) grocery cart C) super market D) fruit basket E) fruit market
Output: The question asks where you put your grapes just before checking out. The answer is the grocery cart. Therefore B.[END OF RESPONSE]

Task 5: Answer Confidence

Prompt for fine-tuned models

''
Please first write analysis of the problem, then select the single correct answer to the following question from the options, and state your option explicitly:
{data}
''

(→ Gets formatted according to model's template)

Example Response

Data: Damage to median nerve produces - A) Claw hand B) Winging ofscapule C) Ape thumb D) Wrist drop
Output: Damage to the median nerve [...] The correct answer is: B) Winging of scapula

Prompt for base models

''